

MEDALLION ARCHITECTURE: THE DATA DESIGN PATTERN POWERING MODERN DATA PLATFORMS



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WHAT IS MEDALLION ARCHITECTURE?

Medallion Architecture is a multi-layered data design pattern that organizes data into progressive stages of quality and refinement.

Instead of transforming raw data directly into reports, data moves through three structured layers
This approach improves scalability, governance, and data quality.

Bronze Layer (Raw Data)



Silver Layer (Clean Data)



Gold Layer (Business Data)

WHY DO WE NEED IT?

Without structured data layers:

**Duplicate
transformations**

Data quality issues

**Difficult
debugging**

**Inconsistent
business metrics**

Poor governance

With Medallion Architecture:

**Single source
of truth**

**Better data
quality**

**Easier
maintenance**

**Faster
analytics**

**Scalable
pipelines**

BRONZE LAYER

**Store data exactly as received
from source systems.**

**Raw and
unprocessed**

**Immutable data
storage**

**Historical records
preserved**

**Supports replay and
auditing**

**Minimal
transformations**

SILVER LAYER

Improve data quality and standardization.

Data cleansing

Deduplication

Type casting

Null handling

Schema validation

Data enrichment

GOLD LAYER

Deliver analytics-ready datasets.

KPIs

Data marts

Business metrics

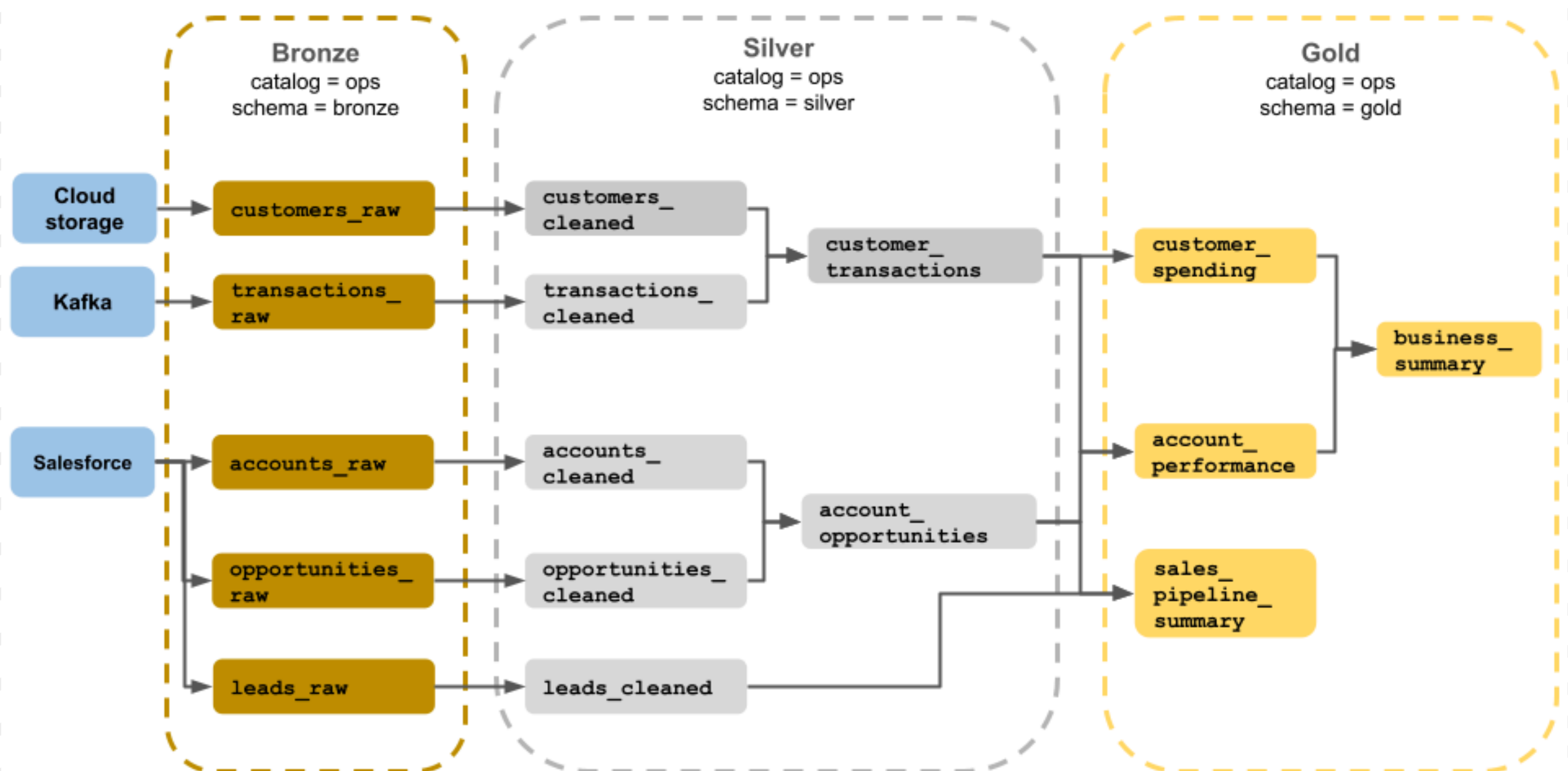
ML feature datasets

Aggregations

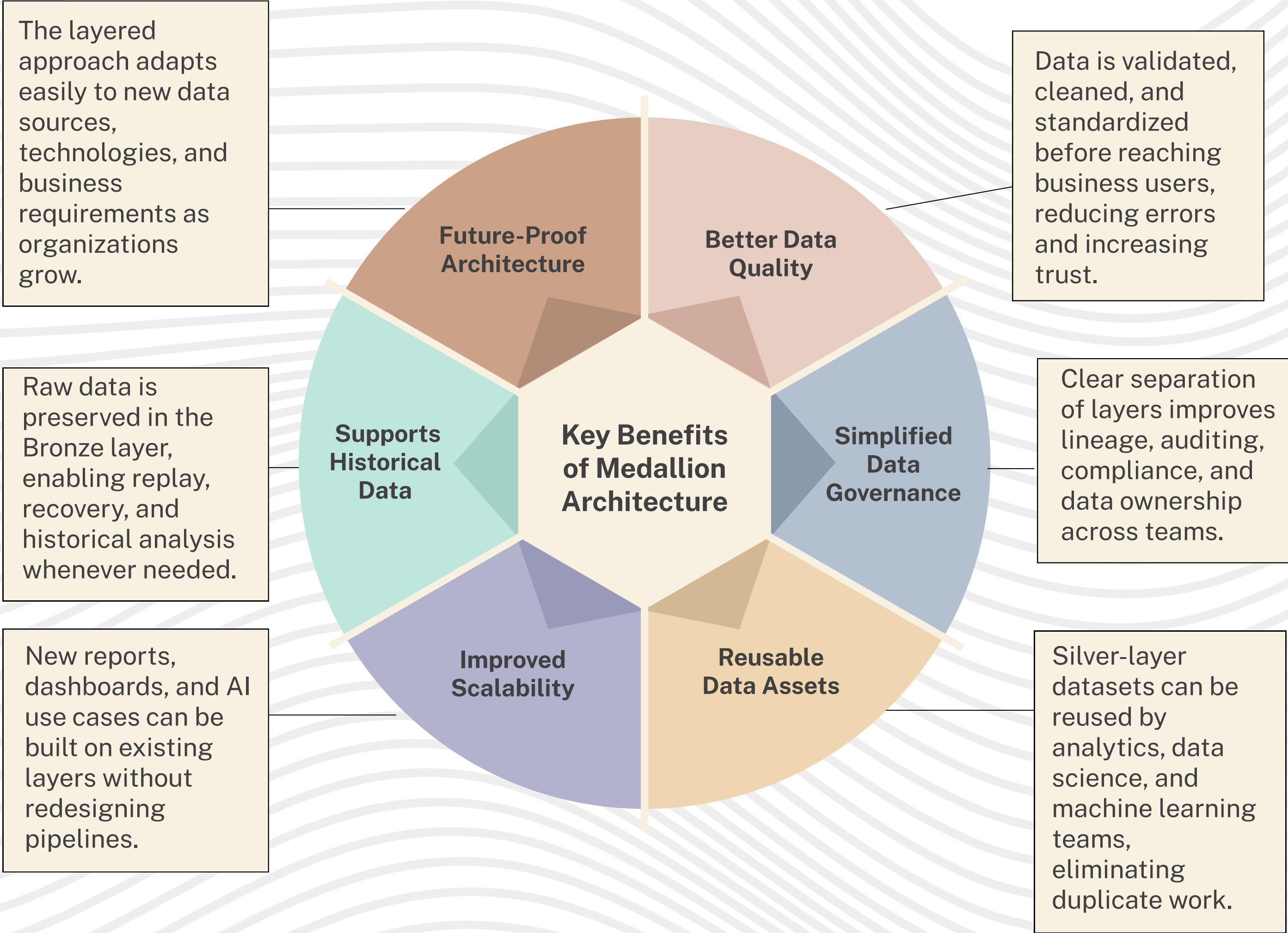
Reporting tables

DATA FLOW EXAMPLE

This example of a medallion architecture shows bronze, silver, and gold layers for use by a business operations team. Each layer is stored in a different schema of the ops catalog.



KEY BENEFITS



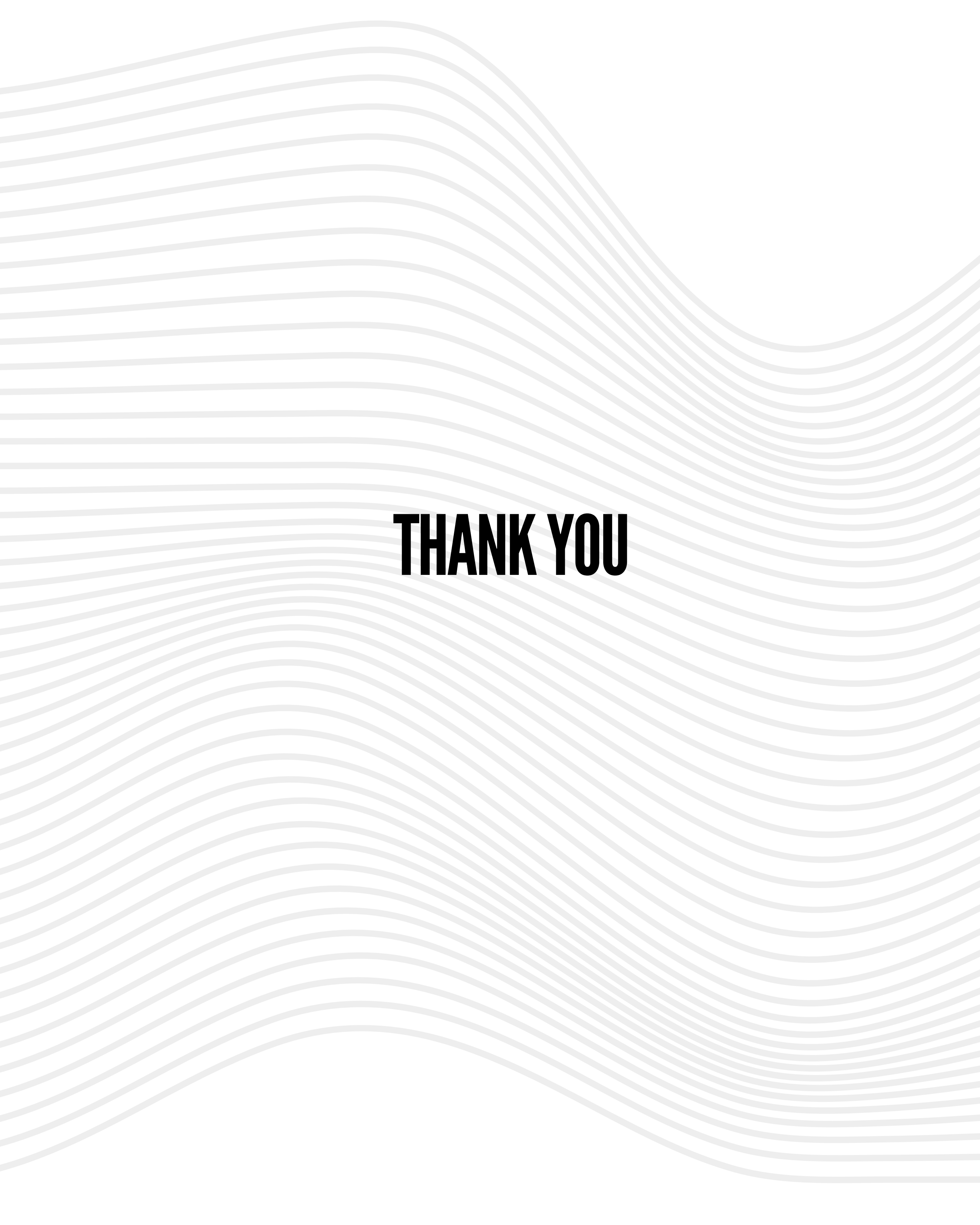
FINAL TAKEAWAY

**THINK OF MEDALLION ARCHITECTURE AS A
DATA REFINERY**



BRONZE = RAW MATERIAL
SILVER = REFINED MATERIAL
GOLD = FINISHED PRODUCT

**THE GOAL ISN'T JUST MOVING DATA.
IT'S PROGRESSIVELY INCREASING DATA
QUALITY, TRUST, AND BUSINESS VALUE AT
EVERY STAGE.**

The background of the image consists of numerous thin, light gray wavy lines that flow across the frame from left to right, creating a sense of movement and depth. These lines are of varying frequencies and amplitudes, giving the background a textured, organic feel.

THANK YOU